

AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 6.3: Krusell–Smith (1998) and DeepHAM

Zhigang Feng

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Topic 6.3 Agenda

1. **Krusell–Smith (1998)** — aggregate uncertainty in HA
2. **DeepHAM** — deep-learning solution

*Arc: T1 Aiyagari + computation → T2a Why ML / Euler → T2b Global DNN → **T3 KS / DeepHAM** → T4 DEQN / OLG → T5 MFG → Lab06*

Krusell–Smith (1998)

Krusell–Smith (1998): aggregate uncertainty

- Aiyagari ignores aggregate fluctuations. Krusell & Smith (1998, *JPE*) add an **aggregate TFP shock** Z_t to the production technology.
- Setup:
 - Continuum of households with idiosyncratic employment $\epsilon \in \{0, 1\}$.
 - Aggregate state $Z_t \in \{Z_g, Z_b\}$, two-state Markov.
 - Joint (ϵ, Z) Markov chain (correlated unemployment and recessions).
 - Inelastic labour, single risk-free bond, borrowing constraint.
- The cross-sectional distribution Γ_t over (a, ϵ) now **moves stochastically over time**.

KS distributions, firms, prices

- Aggregate state: (Z, Γ) where Γ is the joint distribution over individual asset and employment states.
- Firm: $Y_t = Z_t F(K_t, \bar{L})$ with $K_t = \int a d\Gamma_t$ and $\bar{L}$ aggregate labour.
- Factor prices:

$$R_t = Z_t F_K(K_t, \bar{L}) - \delta, \quad w_t = Z_t F_L(K_t, \bar{L}).$$

- Both R_t and w_t depend on the aggregate state through (Z_t, K_t) , which depends on the entire *history* of the economy.

Household problem with the full distribution as state

The HH Bellman equation is now

$$V(a_i, y_i, Z, \Gamma) = \max_{c_i, a'_i} \{ u(c_i) + \beta \mathbb{E}[V(a'_i, y'_i, Z', \Gamma') \mid y_i, Z] \}$$

subject to

$$a'_i = w(Z, K) \bar{\ell} y_i + R(Z, K) a_i - c_i, \quad K = \int a \, d\Gamma,$$

and the law of motion $\Gamma' = G(\Gamma, Z, Z')$.

- **Curse of dimensionality:** Γ is infinite-dimensional.
- Without simplification, the household must *forecast the entire distribution* to make optimal decisions.

State reduction via finite moments

Krusell & Smith (1998) propose: replace Γ with a finite vector of moments

$$m = (m_1, m_2, \dots, m_J), \quad m_j = \int a^j d\Gamma.$$

- Most often: $J = 1$ — only the first moment $m_1 = K = \int a d\Gamma$.
- This dramatically reduces the household state from (a, y, Z, Γ) to (a, y, Z, K) .
- Approximation accuracy is excellent for many calibrations: agents care primarily about *average* capital because that determines r and w .
- “Approximate aggregation” (KS): higher moments rarely matter for the policy.

Postulated log-linear law of motion

Households need to forecast K_{t+1} given (Z_t, K_t) . Postulate

$$\log K_{t+1} = A(Z_t) + B(Z_t) \log K_t,$$

with state-dependent intercept $A(Z)$ and slope $B(Z)$.

- Two unknowns per aggregate state: $\{A(Z_g), B(Z_g), A(Z_b), B(Z_b)\}$.
- Households use this rule to compute $\mathbb{E}[V(a', y', Z', K') \mid y, Z]$.
- In equilibrium, the postulated rule must match the rule *generated* by the household policy under simulation.

KS algorithm: steps 1–2

Step 1. Specify the functional form for the law of motion. Initialise $\{A(Z), B(Z)\}$.

Step 2. Given the guessed law of motion, solve the household DP problem with state (a, y, Z, K) :

$$V(a, y, Z, K) = \max_{a'} \{ u(c) + \beta \mathbb{E}[V(a', y', Z', K'(Z)) \mid y, Z] \}$$

with $K'(Z) = \exp(A(Z) + B(Z) \log K)$.

- This is a 4-dimensional VFI problem (continuous in a and K , discrete in y and Z).
- Standard tools: tensor-product grids, multivariate interpolation, Howard improvement.¹

¹Classical HA computation methods are covered in detail in the video series at <https://space.bilibili.com/2142649036/lists/7180709>.

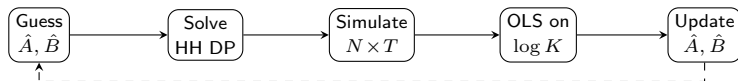
KS algorithm: steps 3–5

Step 3. Simulate the economy under the household policy: $T \approx 10,000$ periods, $N \approx 10,000$ agents. Record the time series $\{K_t, Z_t\}$.

Step 4. Run state-dependent OLS on the simulated $\{K_t, Z_t\}$:

$$\log K_{t+1} = \hat{A}(Z_t) + \hat{B}(Z_t) \log K_t + e_t.$$

Step 5. Update $\{A(Z), B(Z)\} \leftarrow \{\hat{A}(Z), \hat{B}(Z)\}$ and return to Step 2. Iterate until $|A_{\text{new}} - A_{\text{old}}| < \varepsilon$. **Convergence criterion:** in-sample R^2 of the regression close to 1.0 (often > 0.9999 in KS!).



DeepHAM / DEQN replace the OLS step with a neural network trained on simulated states.

Limitations of moment approximation

- Although remarkably accurate *on average*, the moment approximation has weaknesses:
 - *Specific functional form* for the law of motion is hard to choose ex ante.
 - *Higher-order properties* of the distribution (welfare, inequality measures) require more moments.
 - *Optimal-policy analysis* (designing fiscal/monetary intervention) needs derivatives of the value function with respect to the full distribution, not just moments.
- **The natural alternative:** let a neural network *learn* which features of Γ matter.
- This is exactly the DeepHAM / DEQN approach.

DeepHAM

DeepHAM (Han, Yang & E, 2021 preprint, arXiv:2112.14377; Quantitative Economics 17:297–341, 2026)

- DeepHAM solves the (finite- N , large- N) Krusell–Smith problem as a **Markov Nash equilibrium** among N agents.
- Two neural networks:
 - Value function $V(s; \theta^V)$ for individual states.
 - Policy $C(s; \theta^C)$ for individual savings/consumption.
- Training alternates:
 1. **Value step**: simulate under current policy, compute Monte Carlo target, do supervised regression on θ^V .
 2. **Policy step**: solve agent i 's problem with all other agents fixed at the previous policy — *fictitious play*.
- This decomposition makes the high-dimensional joint problem tractable.

DeepHAM: algorithm outline

Initialisation ($k = 0$). Initial policy $\mathcal{C}^{(0)}$, networks $\theta^{V,(0)}, \theta^{C,(0)}$.

k -th iteration:

1. Prepare the stationary distribution $\mu(\mathcal{C}^{(k-1)})$ under the previous policy.
2. Update the value network by supervised learning:

$$\theta^{V,(k)} \in \arg \min_{\theta^V} \mathbb{E}_{\mu} \left[(V(s_i; \theta^V) - \hat{V}_i)^2 \right],$$

with target $\hat{V}_i = \sum_{\tau=0}^{T_v-1} \beta^{\tau} u(c_{\tau}^i)$.

3. Update the policy network by fictitious play:

$$\theta^{C,(k)} \in \arg \max_{\theta^C} \mathbb{E}_{\mu} \left[\sum_{t=0}^{T-1} \beta^t u(c_{i,t}) + \beta^T V(s_{i,T}; \theta^{V,(k)}) \right].$$

DeepHAM: value function learning

- Simulate the economy for T_v periods under the current policy $\mathcal{C}^{(k-1)}$.
- For each sampled initial state s_i , record the realised consumption path and compute the discounted utility target

$$\hat{V}_i = \sum_{\tau=0}^{T_v-1} \beta^\tau u(c_{i,\tau}).$$

- Train $V(\cdot; \theta^V)$ as a regression on $\{(s_i, \hat{V}_i)\}$.
- This is exactly fitted-value Monte Carlo: standard supervised deep learning, no special algorithms required.

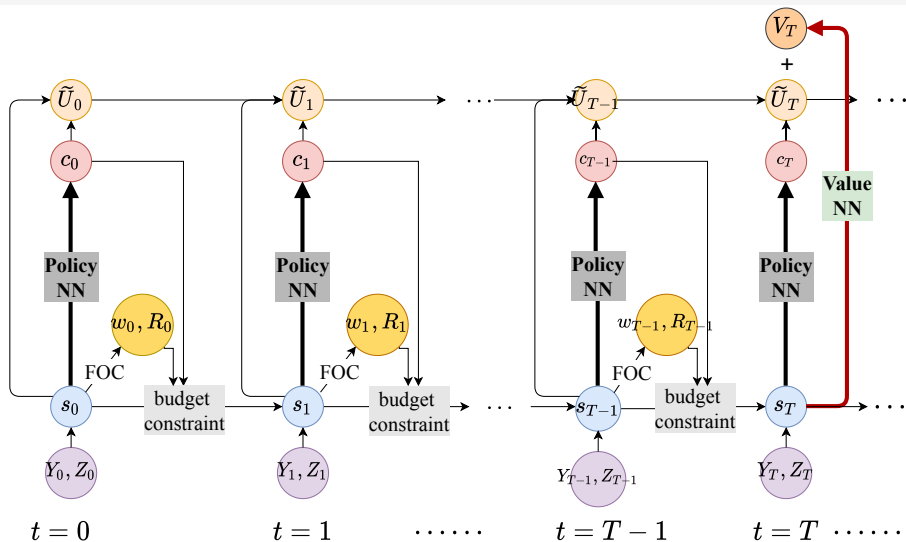
DeepHAM: policy iteration via fictitious play

- In round $\ell + 1$, all agents' policies $\mathcal{C}^{(k,\ell)}$ are known. Agent i improves *best-responding* to the previous policy of all others:

$$\mathcal{C}^{(k,\ell+1)} \in \arg \max_{\mathcal{C}} \mathbb{E}_{\mu} \left[\tilde{U}_{i,T} + \beta^T V^{(k)}(s_{i,T}) \right].$$

- Optimisation is over the parameters of the policy network and runs by gradient ascent on a Monte Carlo sample of paths.
- **Why fictitious play?**
 - Decouples the high-dimensional fixed-point problem.
 - Empirically robust: even in non-convex problems, alternating best response converges to a Nash equilibrium under regularity conditions.

DeepHAM: computational graph



- Inputs: individual state $s = (a_i, y_i, Z, \Gamma)$, with Γ summarised by generalised moments.

DL-based generalised moments

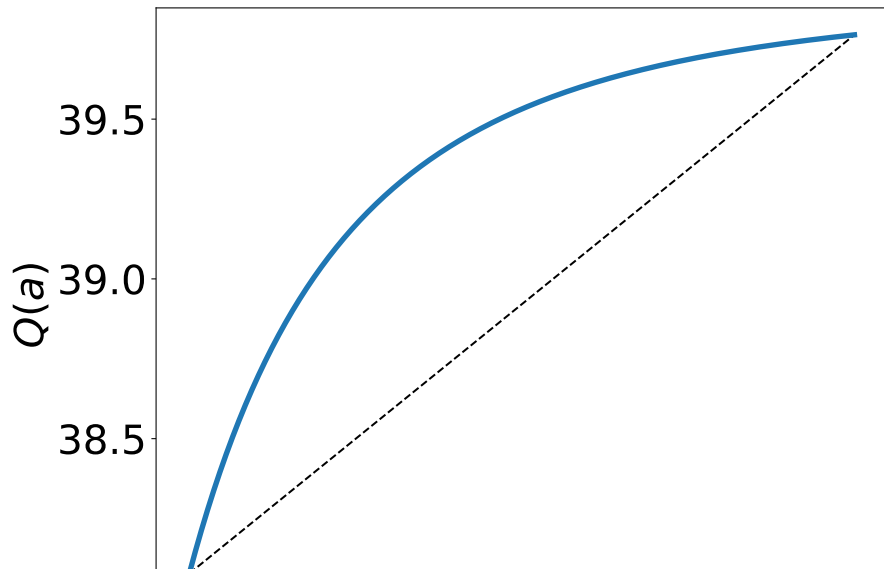
- Instead of a fixed list of moments $\{m_1, m_2, \dots\}$, DeepHAM lets the network *learn* symmetry-preserving sufficient statistics:

$$\bar{Q}_j(\Gamma) = \frac{1}{N} \sum_{i=1}^N Q_j(a_i),$$

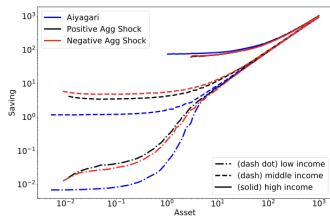
where $Q_1, \dots, Q_J$ are basis functions parameterised by additional NN layers.

- Special case $Q_j(a) = a^j$ recovers the classical moments.
- Trained jointly with V and $\mathcal{C}$, the learned moments adapt to the economic problem at hand.
- Symmetry across agents (the population is exchangeable) is enforced by averaging Q_j across the cross section.

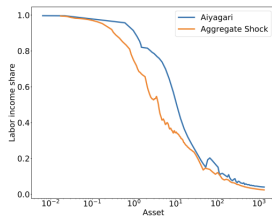
Interpretation of generalised moments



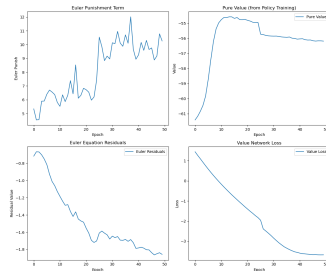
DeepHAM: training diagnostics



(a) Saving policy

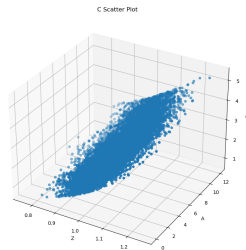
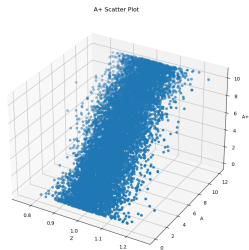


(b) Labor share



- *Left*: simulated cross section of capital under the trained policy.
- *Right*: training loss curves over fictitious-play outer iterations; oscillation reflects the inner-regression vs. outer-policy-improvement decoupling.

DeepHAM: learned policies



- *Left*: learned savings policy, scattered across cross-section samples.
- *Right*: learned consumption policy, scattered across cross-section samples.
- Gap to the classical KS-with-moments benchmark measures the value of DL-based generalised moments.

Next (T4): Deep Equilibrium Nets (a single Euler-loss network) and OLG with aggregate uncertainty.